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RMIT UNIVERSITY

EEET2482 Software Engineering Design
COSC2082 Advanced Programming Techniques
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Assignment Report

AUCTION APPLICATION

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I. INTRODUCTION

Software engineering is an ever-evolving profession-agile, designing, implementing, and maintaining powerful yet efficient systems. This report documents the credit-based auction application developed as part of EEET2482 Software Engineering Design and COSC2082 Advanced Programming Techniques' group assignment. This assignment project, over some time, integrated core concepts of OOP, data structures, memory management, and algorithm design.

This Auction Application has been developed in C++-a console-based system-to emulate real-life scenarios in software development. Primarily designed for its users to engage in buying, selling, and bidding items by using Credit Points (CPs) as the denomination while hosting various functions for registration of

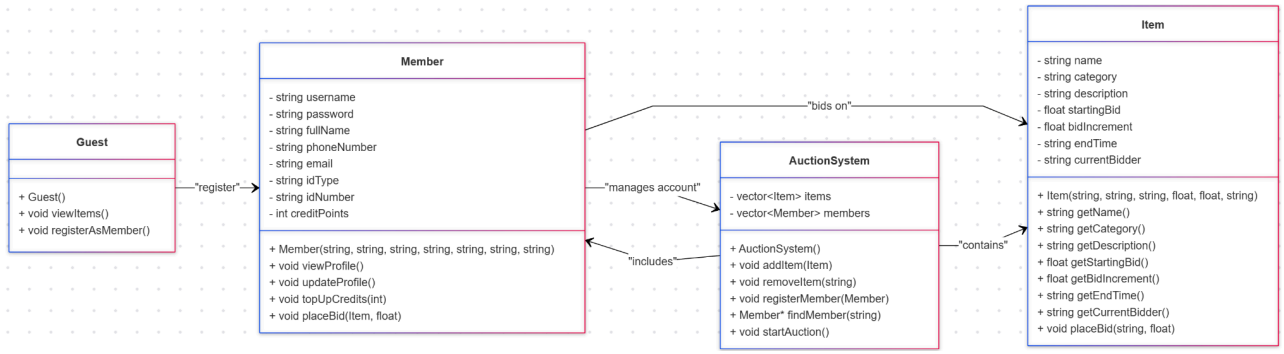
new members, item creation, the bidding process, and profile management. Complementing the aforementioned functionalities with a strong password policy and a role-based access system ensures that user integrity, and therefore the integrity of the application, shall mean business-a good enough reason to invest one's trust in this.

The general purpose of this homework is to provide hands-on experience in translating theory into practice[1] in the framework of professional life. Besides functionality, the application emphasizes scalability, modularity, and maintainability, focusing on competencies important for software development nowadays[2]. This report narrates the design, implementation, and results of the project, casting light upon its architecture and technical issues.

II. APPLICATION DESIGN AND DEVELOPMENT

1. Software Design (Class Diagram)

Class Diagram:



Description of Each Class:

Provide descriptions for each class using the table below:

Class: Guest	Name and Data type	Description	Reason/Explanation
Attribute(s)	No attributes		
Method(s)	Member *signIn(AuctionSystem &auctionSystem);	Prompt username and password to sign in to an account member	Ensures system integrity by allowing only registered members to access member functionalities.
	Member *signInAdmin(AuctionSystem &auctionSystem);	Prompt username and password to sign in to an admin account	
	Member *signUp(AuctionSystem &auctionSystem);	Prompt username, password (ensuring strong policy), and personal info to create and acccount	Allows a guest to create a member account, enforcing strong password policies to ensure security.
	void limitedItemView(AuctionSystem &auctionSystem);	Get item keyword to view existing items with limited info (name, category, description)	Enables guests to view item details with limited information to incentivize registration.

Class: Member	Name and Data type	Description	Reason/Explanation
Attribute(s)	std::string username;	username	personal info
	std::string password;	password	personal info
	std::string fullName;	full name	personal info
	std::string phonenumber;	phonenumber	personal info
	std::string email;	email	personal info
	std::string IDtype;	id type	personal info
	std::string IDnumber;	id number	personal info
	int isAdmin;	set a flag to define whether a member is admin or not	same as description
	int itemsWonToDate;	number of items won	same as description
	int CP;	credit point	personal info
	float rating;	rating	personal info
	std::vector<Item *> activeItemListings;	list of active items placed for bid	keep track of items that the person is selling
	std::unordered_map<Item *, int> activeBids;	map the bid value of member to an item	keep track of bid across members
	std::vector<std::string > reviewsByOthers;	reviews from others	same as description
Method(s)	setters / getters function	modify member attribute	modify member attribute
	void viewDashBoard();	Members can view their active item listing, all active bids, number of items won, credit points, their rating, reviews left by others as a single dashboard	view personal account
	void topUpCredit();	Top up credit but member need to enter password -> ensuring authority	top up credit
	void changePassword(AuctionSystem &auctionSystem);	Change password with password policy	change password
	void changeUsername(AuctionSystem &auctionSystem);	change username ensuring that username across all members are unique	change username
	void proItemSearch(AuctionSystem &auctionSystem);	search for item by keywords, CP range	search items with advance feature
	void placeBid(AuctionSystem &auctionSystem);	Create a new item on active item listing	creat new item and add to active item listing

	void Bid(AuctionSystem &auctionSystem);	Prompt item ID to bid on	bid on an item
	void removeBid(AuctionSystem &auctionSystem);	Remove bid from active item listing, ensuring that theres no one current bid on that item	remove bid from active bids
	void setItemRating();	Set required rating for an item on active item listing	same as description
	void reviewOther(AuctionSystem &auctionSystem);	Review other people	review other
	void showReviewsByOther();	Show reviews left by others	list of reviews from other
	void viewAnyProfile(AuctionSystem &auctionSystem);	This method only works when isAdmin flag is set to 1, admin can view any items detail	admin privilege

Class: Item	Name and Data type	Description	Reason/Explanation
Attribute(s)	std::string itemName;	Name of item	Basic information for identifying and categorizing the item.
	std::string category;	Category of item	
	std::string description;	Description of item	
	std::string endTime;	Item end date bid	
	std::string currentBidder;	Current bidder hwo has the highest bid on the item	keep track of bidder with highest bid
	std::vector<Member *> bidders;	List of bidders	Maintains a list of members bidding on the item, allowing transparency and competition.
	Member *owner;	Item owner	keep track of the owner
	int bidIncrement;	Bid increment for the item	bid increment for bid requirement, e.g. new bid must larger than current with x credits (with x = bidIncrement)
	float ratingScore;	Rating score required for the item	set rating score required to view item
	int itemId;	Item Id	this variable is unique so that the system can access it via Id
	int highestBid;	Current highest bid on the item	basic item info
Method(s)	setters / getters function	modify item's attribute	same as description
	void addBidders(Member *member);	Add new bidders	add members who bid on the item

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Class: AuctionSystem	Name and Data type	Description	Reason/Explanation
Attribute(s)	vector<Member*> members	Hold list of members in the Auction Sytem	Centralize members
	vector<Item*> items	Hold list of items in Auction System	Centralize items
Method(s)	void addMember(Member *member);	Add member (registered account) to system	Manage the addition of members and items to the system.
	void addItem(Item *item);	Add item to system	
	void removeItem(Item*item);	Remove item from system	Manage the removal of members and items from the system.
	void turnAdmin(Member* member)	set isAdmin of member to 1	give admin privilege to member
	bool isUsernameTaken(const std::string &username);	Check whether username is taken or not	Ensure uniqueness in username
	bool isValidPassword(const std::string &password);	Ensure password policy	Support password policy
	bool checkUserName(std::string username);	Check whether a user is existed	same as description

Class Relationships:

- **Guest to AuctionSystem:** Dependency. Guests interact with the AuctionSystem to view items and register.
- **Member to AuctionSystem:** Composition. Members are part of the AuctionSystem and interact with items and other members., Delete auction system, member account also deleted
- **Item to Member:** Association. Items have an owner, represented by a Member object. Member has a list of active item listing and active bids of items.
- **AuctionSystem to Item:** Aggregation. The AuctionSystem manages a collection of items.

2. Implementation Result

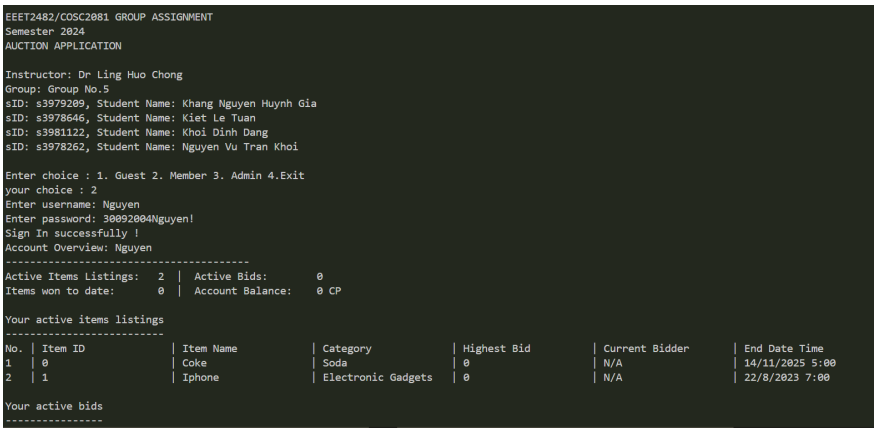
a. Welcome Screen

Result Summary:

The welcome of the Auction Application introduces the system in order and navigates a user through the functionality of the system. Once opened, it gives information on course and group, instructor's name, Group details, and assignment title. It then asks to select one of the given roles: Guest, Member, and Admin. Depending upon the role it gives them different options to enter into the system. Member, upon successful login, is taken to their dashboard page which shows the overview of

the account, Active Items Listed and Bids with crucial details including Credit Balance and Auction End Times. This will make the navigation easier and users will find it convenient.

Screenshots of Sample Result:



b. Basic Features

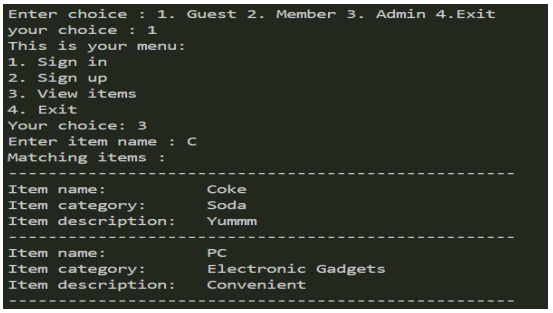
Result Summary:

Feature Name	Feature Description	Status (*Implemented/ Not Implemented*)	Bugs/Limitations if it has
1. Non-member access	Non-member (as guest) can view item listings with limited details: item name, category, and short description.	implemented	none
2. Non-member registration / sign up	Non-member (as guest) can register to become a member. Upon registration, they will start with 0 credit points and a default rating of 3 (all information is recorded and stored). The registration process must include a strong password policy.	implemented	none
3. Non-member limited item view	Non-member (as guest) can search item with limited info: item name, category, description	implemented	none
4. Member log in	Members can log in using username and password	Implemented	none
5. Member profile management	Members can view and edit their profiles, including rating view, show reviews by others, top up credit points and change password.	Implemented	none
6. Active item listing creation	Members can list items for auction with details like category, required rating, end date, . . .	implemented	none

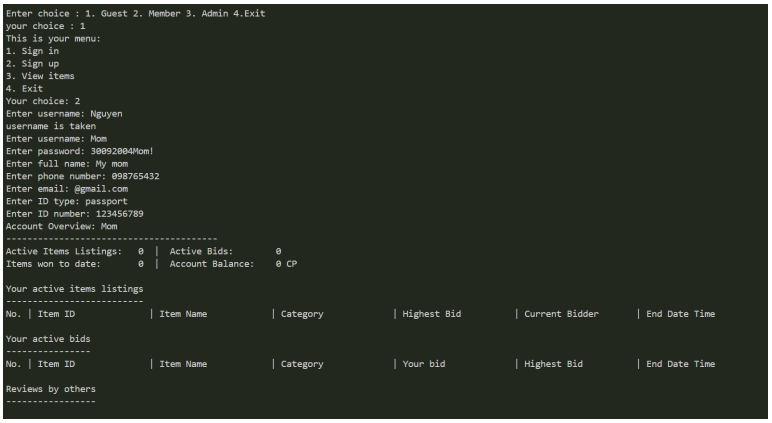
7. Advance item search	Member can search items by name, category, or credit point range and view details like current highest bid, the current bidder and the auction's end date & time.	implemented	none
8. Bid functionality	Member can place bid if they meet CP requirement	implemented	none
9. Handle transaction	Transaction happens everytime a member sign in. To ensure it fits end date & time of items. Member can perform ratings, reviews after completing a transaction, CP point updated for buyer and seller after transaction	implemented	none
10. Admin privilege	View any profile	implemented	none

Screenshots of Sample Result (for each feature):

View item listings with limited details:



Non member registration:



3. Non member limited item view:

```
Enter choice : 1. Guest 2. Member 3. Admin 4.Exit
your choice : 1
This is your menu:
1. Sign in
2. Sign up
3. View items
4. Exit
Your choice: 3
Enter item name : C
Matching items :
-----------------------------------------------------
Item name:          Coke
Item category:      Soda
Item description:   Yummm
-----------------------------------------------------
Item name:          PC
Item category:      Electronic Gadgets
Item description:   Convenient
-----------------------------------------------------
```

4. Member log in

```
Enter choice : 1. Guest 2. Member 3. Admin 4.Exit
your choice : 2
Enter username: Nguyen
Enter password: 30092004Nguyen!
Sign In successfully !
Account Overview: Nguyen
---------------------------------------
Active Items Listings:   2  |  Active Bids:       0
Items won to date:       0  |  Account Balance:   0 CP

Your active items listings
--------------------------
No. | Item ID             | Item Name           | Category            | Highest Bid         | Current Bidder      | End Date Time
1   | 0                   | Coke                | Soda                | 0                   | N/A                 | 14/11/2025 5:00
2   | 1                   | Iphone              | Electronic Gadgets  | 0                   | N/A                 | 22/8/2023 7:00

Your active bids
----------------
No. | Item ID             | Item Name           | Category            | Your bid            | Highest Bid         | End Date Time

Reviews by others
-----------------

My rating
---------
Rating: 3
```

5. Member profile management

+ change password:

```
This is your menu:
1.  View dashboard
2.  Top up credit
3.  Change password
4.  Change username
5.  Search items
6.  Place bid
7.  Bid
8.  Remove bid
9.  Set item rating
10. Review other member
11. Show others review
12. Exit
Your choice: 3
Enter new password : 123
Password must be at least 8 characters long
Enter new password : 123Errrrrrr
Password must contain at least one special character
Enter new password : 123Erik!
```

+ topping up credit:

```
This is your menu:
1.  View dashboard
2.  Top up credit
3.  Change password
4.  Change username
5.  Search items
6.  Place bid
7.  Bid
8.  Remove bid
9.  Set item rating
10. Review other member
11. Show others review
12. Exit
Your choice: 2
Enter password : 890Erik!
Enter credit to top up: 9000
```

+ (Many more . . .)

6. Add item to active item list:

```
This is your menu:
1.  View dashboard
2.  Top up credit
3.  Change password
4.  Change username
5.  Search items
6.  Place bid
7.  Bid
8.  Remove bid
9.  Set item rating
10. Review other member
11. Show others review
12. Exit
Your choice: 6
Enter item name: Toy
Enter end date time: 17/01/2025
Enter category: Toy
Enter description: Fun
```

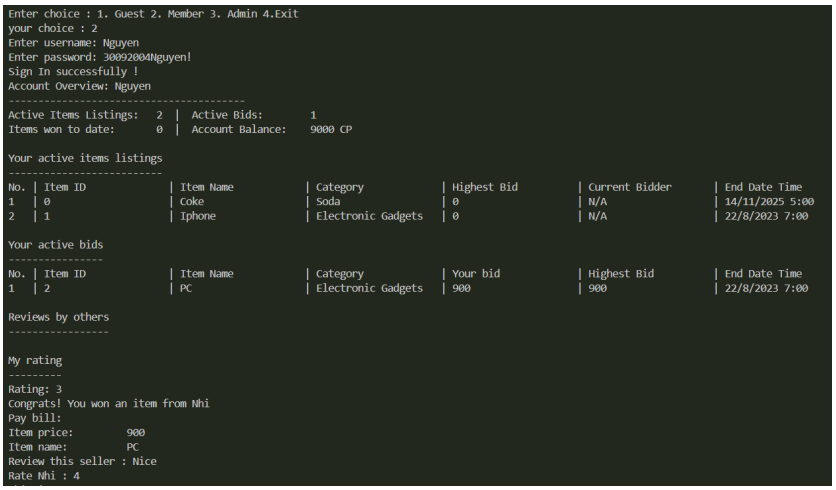
7. Advance Item search

```
Enter item name : Coke
Enter category : S
Enter minimum CP : 0
Enter maximum CP : 100
Matching Items :
---------------------------------------------
Name:               Coke
Category:           Soda
Description:        Yummm
End Date Time:      14/11/2025 5:00
Highest Bid:        0
Current Bidder:     N/A
Item Id:            0
---------------------------------------------
```

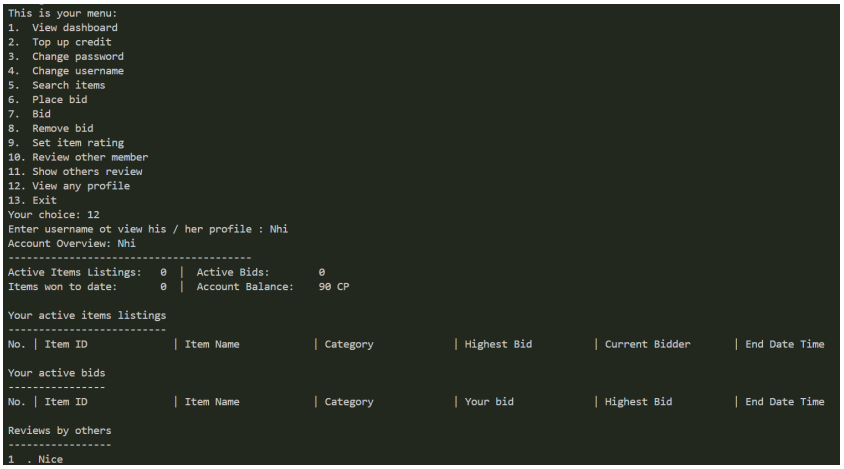
8 . Bid functionality

```
This is your menu:
1.  View dashboard
2.  Top up credit
3.  Change password
4.  Change username
5.  Search items
6.  Place bid
7.  Bid
8.  Remove bid
9.  Set item rating
10. Review other member
11. Show others review
12. Exit
Your choice: 7
Enter item ID to bid on : 3
Item detail:
---------------------------------------------
Name:               Toy
Category:           Toy
Description:        Fun
End Date Time:      17/01/2025
Highest Bid:        0
Current Bidder:     N/A
Item Id:            3
---------------------------------------------
Enter your bid : 5
Your bid must be as least 20 CP higher than current bid (0)
```

9. Handle transaction:



10. View any profile from admin privilege:



c. Advanced Features

Result Summary:

Feature Name	Feature Description	Status (*Implemented/ Not Implemented*)	Bugs/Limitations if it has
1. Member dashboard	Members can view their active item listing, all active bids, number of items won, credit points, their rating, reviews left by others as a single dashboard	Implemented	None
2. Bid Limit Automation	User can bid on an item, the program will show warning if his / her CP is unaffordable, the bid must be higher than the current bid + bid increment of the item, bid limit feature has not yet been developed	Not Implemented	To be addressed in future work

Screenshots of Sample Result:

Dashboard:



IV. DISCUSSIONS & CONCLUSIONS

Overall, it can be elicited that the realization of the project has met its objectives and designed principles as were set out in the design. Core functionalities, such as user registration, item bidding, and profile management, amongst others, are well implemented to represent the relationship between classes outlined in the design phase through the object-oriented programming principles. These ensure modularity and maintainability within the auction application.

Key Takeaways

+ Development Efficiency: Design brought clarity and structure; therefore, there was nothing to redefine or rework, just efficient coding.
+ Practical Changes: Iterative updating, promoted by practical difficulties, introduced improved functionality without the loss of the original aims.

Lessons Learned

+ Technical Skills: Stronger knowledge about programming with C++, particularly the application of advanced concepts such as memory management in general and data structures.
+ Experience in implementing security measures, such as password policies and access control have been gained.
+ Team Collaboration: Improved teamwork ability and project management skills on how tasks can be properly distributed and controlled through version control.

This assignment filled in many of the gaps between theory and practice regarding how software development processes are executed. My skills and experience developed here will be helpful in forming a strong base for future works at both an academic and professional level.

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V. REFERENCES (USE IEEE STYLES)

[1] Savin-Baden, M., & Major, C. (2023). *Qualitative research: The essential guide to theory and practice*. Routledge.

[2] Adnan, N. H., & Ritzhaupt, A. D. (2018). Software engineering design principles applied to instructional design: What can we learn from our sister discipline?. *TechTrends*, *62*(1), 77-94.